

Closed-Circuit

Detection and automated isolation of broken Low-Voltage AC overhead distribution conductors.

Smart India Hackathon 2026 (9th edition) · Team ID

VITBSIH26-388

Theme: Disaster Management · Category: Hardware ·

Problem source: KSEBL, Govt. of Kerala

Mentor: Dr. Abha Trivedi, SCAI, VIT Bhopal University

1. The problem in one paragraph

When an LV overhead conductor snaps, the live end can stay energised on the ground for minutes to hours. Conventional protection (fuses, overcurrent relays) frequently does not see a broken conductor at all, because a wire lying on dry earth or asphalt draws far less than the trip current. The result is electrocution risk and fire risk with no automatic detection. Existing solutions require either expensive line-mounted CTs or full smart-meter rollout.

2. Our approach

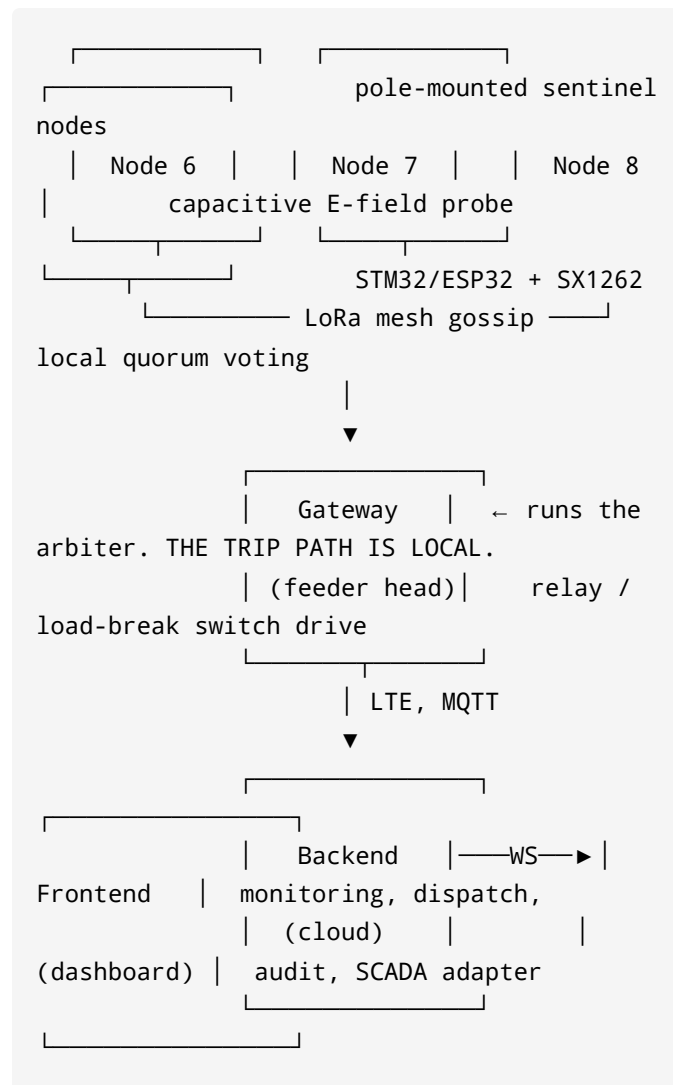
A live conductor radiates a 50 Hz electric field. A cheap non-contact capacitive probe mounted on the pole, a few tens of centimetres from the line, can sense that field without any galvanic connection, no CT, no line tap, no outage to install.

When a conductor breaks, the E-field downstream of the break **collapses**. A single node seeing a collapse is not trustworthy – rain, fog, vegetation contact and switching transients all perturb the field. So nodes gossip over a LoRa mesh and a **quorum** of neighbours must agree before anything is asserted. Only then does a gateway at the feeder head command isolation.

Design targets

Metric	Target
Detection to isolation	< 2 s
False trip rate	0 in bench characterisation set
Node BOM cost	< ₹1,800
Node average current	< 5 mA (solar + LiFePO4, no mains)
Inter-node range	≥ 300 m LoRa, line of sight along span

3. Architecture



Non-negotiable architectural rule: the isolation decision executes on the gateway, not in the cloud.

LTE round-trip alone can blow the 2 s budget, and a system whose safety function depends on an internet connection is unsellable to a utility. The cloud observes, alerts, audits and configures. It does not trip.

4. Repository layout

```
closed-circuit/  
├─ README.md          ← you are here  
├─ docs/  
│   ├─ PROTOCOL.md    ← THE SHARED  
│   │                   CONTRACT. Read before writing any code.  
│   └─ CONTRIBUTING.md ← branching,  
│                   reviews, definition of done  
├─ simulator/         ← virtual feeder,  
│                   emits protocol-conformant telemetry  
├─ backend/           ← FastAPI ingest,  
│                   arbiter, WebSocket fan-out, REST  
├─ frontend/          ← React  
│                   dashboard, map, fault cascade view  
├─ gateway/           ← LoRa  
│                   concentrator service + relay actuation  
├─ firmware/          ← node firmware,  
│                   sampling, detector, LoRa mesh  
└─ hardware/          ← AFE schematic,  
│                   PCB, enclosure, characterisation data
```

Each directory has its own README with a task list, acceptance criteria and gotchas. **Start with** `docs/PROTOCOL.md`.

5. Ownership

Six people, six tracks. Fill in the names.

Track	Directory	Owner	Notes
Hardware / AFE	hardware/	TBD	Starts today. PCB lead time is the critical path.
Firmware / mesh	firmware/	TBD	Works on dev boards until PCB arrives.
Gateway / actuation	gateway/	TBD	Owens the safety-critical trip path.
Simulator	simulator/	TBD	Unblocks backend + frontend on day 1.
Backend	backend/	TBD	Owens <code>decide()</code> , which the gateway

Track	Directory	Owner	Notes
			later reuses.
Frontend	frontend/	TBD	Owens the demo.

Pranav (team lead) – integration, PPT, demo script, PROTOCOL ownership.

6. Milestones

#	Milestone	Definition of done
M0	Contract frozen	docs/PROTOCOL .md merged, all six people have read it
M1	Demo link live	Public URL, simulated feeder, scenario buttons, cascade animation
M2	AFE validated	Bench characterisation dataset: standoff, temperature, humidity
M3	Three-node mesh	Real LoRa gossip between 3 dev boards, quorum fires correctly

#	Milestone	Definition of done
M4	Gateway trips a relay	Physical relay clicks from a real quorum event, < 2 s measured
M5	Scaled field test	Mock pole span, energised conductor, physical cut, video recorded
M6	Cloud integration	Gateway → MQTT → backend → dashboard, end to end

M1 is what goes in the PPT. M5 is what wins the finale.

7. Demo honesty

The submitted demo runs on a **simulated feeder with a live consensus engine**. Label it that way in the PPT and say it out loud during the demo. If a judge works out the data is synthetic after you have implied otherwise, you lose the room. Stated upfront, it reads as competent engineering practice — which it is.

8. Quick start (demo stack)

```
git clone <repo> && cd closed-circuit

# backend + simulator
cd backend && python -m venv .venv &&
source .venv/bin/activate
pip install -r requirements.txt && uvicorn
app.main:app --reload

# frontend, separate terminal
cd frontend && npm install && npm run dev
```

Open <http://localhost:5173>. The simulator runs inside the backend process — no third terminal, no broker needed locally.